Two Dimensional Binary Indexed Tree or Fenwick Tree

Last Updated : 30 Nov, 2023

Prerequisite - Fenwick Tree

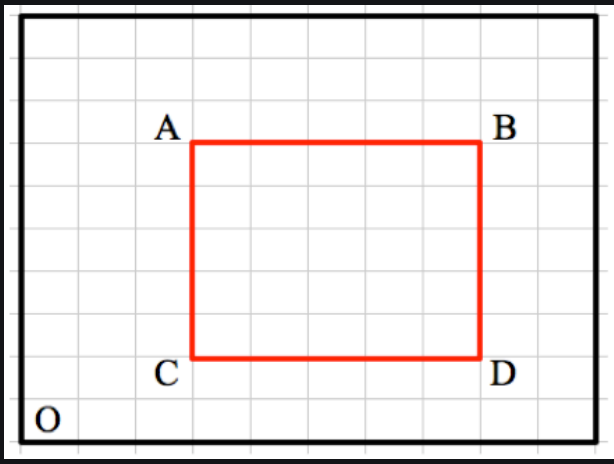
We know that to answer range sum queries on a 1-D array efficiently, binary indexed tree (or Fenwick Tree) is the best choice (even better than segment tree due to less memory requirements and a little faster than segment tree).

Can we answer sub-matrix sum queries efficiently using Binary Indexed Tree ?

The answer is **yes**. This is possible using a **2D BIT** which is nothing but an array of 1D BIT.

Algorithm:

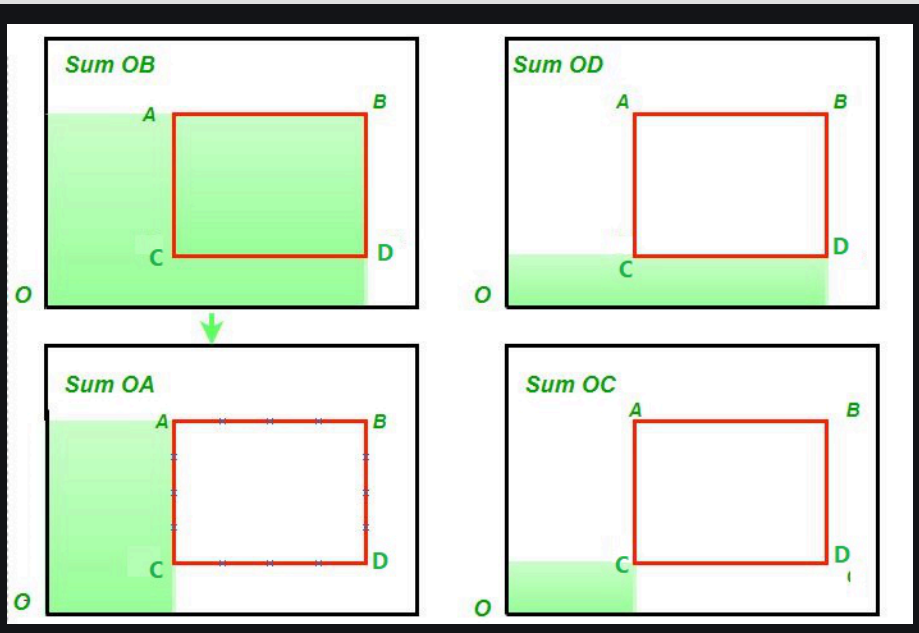
We consider the below example. Suppose we have to find the sum of all numbers inside the highlighted area:



70	37	23	57	27	22	90	99	22	59
47	63	33	1	42	46	6	70	98	93
36	62	50	21	92	27	60	29	15	34
53	3	88	45	57	39	83	81	79	56
28	63	89	20	47	15	84	18	82	33
26	87	11	76	79	5	94	55	73	51
17	82	86	10	96	5	42	43	51	6
44	76	51	4	15	99	52	11	70	89
66	36	92	85	50	21	72	27	52	65
60	0	67	37	59	14	33	13	36	36

We assume the origin of the matrix at the bottom - O.Then a 2D BIT exploits the fact that-

```
Sum under the marked area = Sum(OB) - Sum(OD) -
                            Sum(OA) + Sum(OC)
```



In our program, we use the getSum(x, y) function which finds the sum of the matrix from (0, 0) to (x, y).
Hence the below formula :

```
Sum under the marked area = Sum(OB) - Sum(OD) -
                            Sum(OA) + Sum(OC)

The above formula gets reduced to,

Query(x1,y1,x2,y2) = getSum(x2, y2) -
                     getSum(x2, y1-1) -
                     getSum(x1-1, y2) +
                     getSum(x1-1, y1-1)
```

where,
x1, y1 = x and y coordinates of C
x2, y2 = x and y coordinates of B
The updateBIT(x, y, val) function updates all the elements under the

region – (x, y) to (N, M) where,

N = maximum X co-ordinate of the whole matrix.

M = maximum Y co-ordinate of the whole matrix.

The rest procedure is quite similar to that of 1D Binary Indexed Tree.

Below is the C++ implementation of 2D indexed tree

C++ Java Python3 C# JavaScript

```
/* C++ program to implement 2D Binary Indexed Tree

2D BIT is basically a BIT where each element is another BIT.
Updating by adding v on (x, y) means it's effect will be found
throughout the rectangle [(x, y), (max_x, max_y)],
and query for (x, y) gives you the result of the rectangle
[(0, 0), (x, y)], assuming the total rectangle is
[(0, 0), (max_x, max_y)]. So when you query and update on
this BIT,you have to be careful about how many times you are
subtracting a rectangle and adding it. Simple set union formula
works here.

So if you want to get the result of a specific rectangle
[(x1, y1), (x2, y2)], the following steps are necessary:

Query(x1,y1,x2,y2) = getSum(x2, y2)-getSum(x2, y1-1) -
                     getSum(x1-1, y2)+getSum(x1-1, y1-1)

Here 'Query(x1,y1,x2,y2)' means the sum of elements enclosed
in the rectangle with bottom-left corner's co-ordinates
(x1, y1) and top-right corner's co-ordinates - (x2, y2)

Constraints -> x1<=x2 and y1<=y2

    /\
 y  |
    |          --------(x2,y2)
    |         |       |
    |         |       |
    |         |       |
    |         ---------
```

```
    |       (x1,y1)
    |
    |_____________________________
   (0, 0)                    x-->

In this program we have assumed a square matrix. The
program can be easily extended to a rectangular one. */

#include<bits/stdc++.h>
using namespace std;

#define N 4 // N-->max_x and max_y

// A structure to hold the queries
struct Query
{
    int x1, y1; // x and y co-ordinates of bottom left
    int x2, y2; // x and y co-ordinates of top right
};

// A function to update the 2D BIT
void updateBIT(int BIT[][N+1], int x, int y, int val)
{
    for (; x <= N; x += (x & -x))
    {
        // This loop update all the 1D BIT inside the
        // array of 1D BIT = BIT[x]
        for (int yy=y; yy <= N; yy += (yy & -yy))
            BIT[x][yy] += val;
    }
    return;
}

// A function to get sum from (0, 0) to (x, y)
int getSum(int BIT[][N+1], int x, int y)
{
    int sum = 0;

    for(; x > 0; x -= x&-x)
    {
```

```
        // This loop sum through all the 1D BIT
        // inside the array of 1D BIT = BIT[x]
        for(int yy=y; yy > 0; yy -= yy&-yy)
        {
            sum += BIT[x][yy];
        }
    }
    return sum;
}

// A function to create an auxiliary matrix
// from the given input matrix
void constructAux(int mat[][N], int aux[][N+1])
{
    // Initialise Auxiliary array to 0
    for (int i=0; i<=N; i++)
        for (int j=0; j<=N; j++)
            aux[i][j] = 0;

    // Construct the Auxiliary Matrix
    for (int j=1; j<=N; j++)
        for (int i=1; i<=N; i++)
            aux[i][j] = mat[N-j][i-1];

    return;
}

// A function to construct a 2D BIT
void construct2DBIT(int mat[][N], int BIT[][N+1])
{
    // Create an auxiliary matrix
    int aux[N+1][N+1];
    constructAux(mat, aux);

    // Initialise the BIT to 0
    for (int i=1; i<=N; i++)
        for (int j=1; j<=N; j++)
            BIT[i][j] = 0;

    for (int j=1; j<=N; j++)
```

```
    {
        for (int i=1; i<=N; i++)
        {
            // Creating a 2D-BIT using update function
            // everytime we/ encounter a value in the
            // input 2D-array
            int v1 = getSum(BIT, i, j);
            int v2 = getSum(BIT, i, j-1);
            int v3 = getSum(BIT, i-1, j-1);
            int v4 = getSum(BIT, i-1, j);

            // Assigning a value to a particular element
            // of 2D BIT
            updateBIT(BIT, i, j, aux[i][j]-(v1-v2-v4+v3));
        }
    }

    return;
}

// A function to answer the queries
void answerQueries(Query q[], int m, int BIT[][N+1])
```

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```
    for (int i=0; i<m; i++)
    {
        int x1 = q[i].x1 + 1;
        int y1 = q[i].y1 + 1;
        int x2 = q[i].x2 + 1;
        int y2 = q[i].y2 + 1;

        int ans = getSum(BIT, x2, y2)-getSum(BIT, x2, y1-1)-
                  getSum(BIT, x1-1, y2)+getSum(BIT, x1-1, y1-
1);

        printf ("Query(%d, %d, %d, %d) = %d\n",
                q[i].x1, q[i].y1, q[i].x2, q[i].y2, ans);
    }
    return;
}

// Driver program
```

```
int main()
{
    int mat[N][N] = {{1, 2, 3, 4},
                     {5, 3, 8, 1},
                     {4, 6, 7, 5},
                     {2, 4, 8, 9}};

    // Create a 2D Binary Indexed Tree
    int BIT[N+1][N+1];
    construct2DBIT(mat, BIT);

    /* Queries of the form - x1, y1, x2, y2
       For example the query- {1, 1, 3, 2} means the sub-
matrix-
    y
    /\
 3  |       1 2 3 4      Sub-matrix
 2  |       5 3 8 1      {1,1,3,2}      --->     3 8 1
 1  |       4 6 7 5                                6 7 5
 0  |       2 4 8 9
    |
  --|------ 0 1 2 3 ----> x
    |

    Hence sum of the sub-matrix = 3+8+1+6+7+5 = 30

    */

    Query q[] = {{1, 1, 3, 2}, {2, 3, 3, 3}, {1, 1, 1, 1}};
    int m = sizeof(q)/sizeof(q[0]);

    answerQueries(q, m, BIT);

    return(0);
}
```

Output

```
Query(1, 1, 3, 2) = 30
Query(2, 3, 3, 3) = 7
```

```
Query(1, 1, 1, 1) = 6
```

Time Complexity:

- Both updateBIT(x, y, val) function and getSum(x, y) function takes O(log(N)*log(M)) time.
- Building the 2D BIT takes O(NM log(N)*log(M)).
- Since in each of the queries we are calling getSum(x, y) function so answering all the Q queries takes **O(Q*log(N)*log(M))** time.
- Hence the overall time complexity of the program is **O((NM+Q)*log(N)*log(M))** where,
  N = maximum X co-ordinate of the whole matrix.
  M = maximum Y co-ordinate of the whole matrix.
  Q = Number of queries.

Auxiliary Space: O(NM) to store the BIT and the auxiliary array

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